Chapter 7

Potentiometry and Dead-Stop Titration

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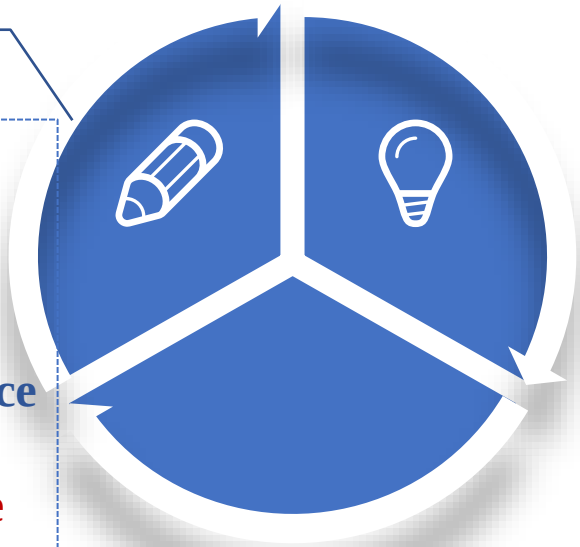
Contents

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- Section 2 Basic Principles of Potentiometry
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Learning Objectives

Mast

- Basic principles of **direct potentiometry**; methods and precautions for **pH measurement** by direct potentiometry; significance and role of the **selectivity coefficient** of **ion-selective electrodes**; methods for determining endpoints in potentiometric titration and **dead-stop titration**



掌握

- **直接电位法**的基本原理；直接电位法**测定溶液pH**的方法及注意事项；**离子选择电极**的**选择性系数**的意义、作用；电位滴定法和**永停滴定法**确定终点的方法

Learning Objectives



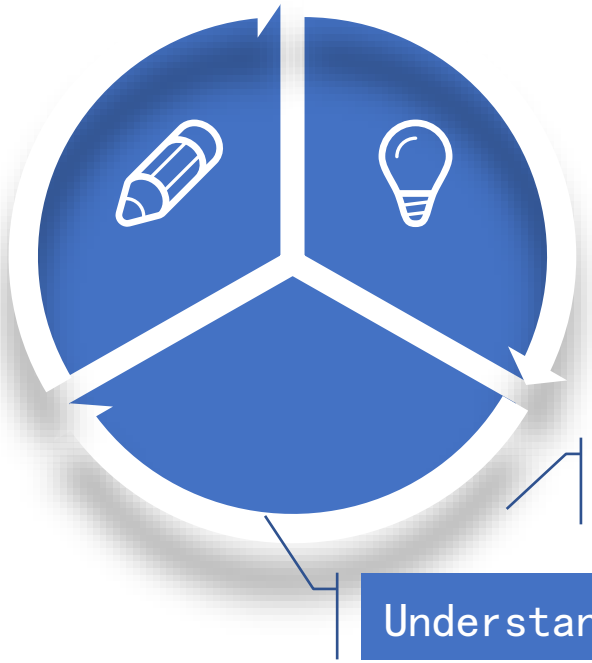
Familiar with

- Common **indicator and reference electrodes**; structure, response mechanism, and performance of the **pH glass electrode**; methods for determining other **cation**/ˈkætaɪən/ and **anion**/ˈænaɪən/ concentrations and the role of TISAB

熟悉

- 常用的**指示电极和参比电极**；**pH玻璃电极**的结构、响应机制、性能；其他**阴、阳离子**浓度的测定方法及TISAB的作用

Learning Objectives



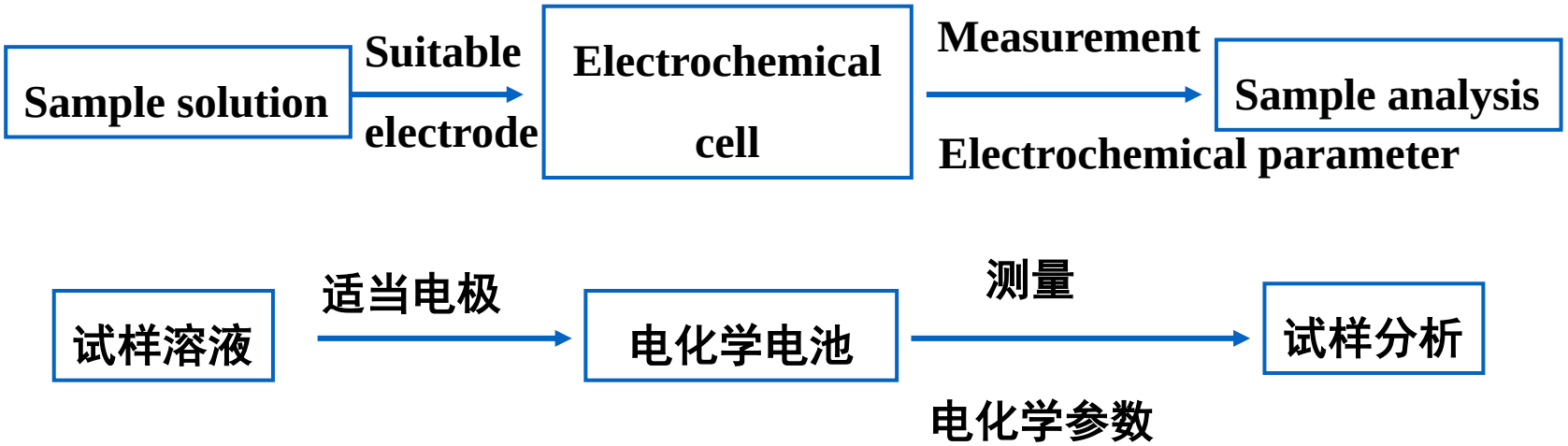
- 电化学分析法及其分类；相界电位、液接电位、膜电位、不对称电位；复合pH电极；离子选择电极的分类

- **Electrochemical analysis and its classification; phase-boundary potential, liquid-junction potential, membrane potential, asymmetry potential; composite pH electrode; classification of ion-selective electrodes**

Section 1 Overview

Section 1 Overview

Electrochemical analysis电化学分析or electroanalytical chemistry电分析化学uses a sample solution and suitable electrodes to form an electrochemical cell电化学电池. It analyzes a sample by measuring the intensity or change of electrochemical parameters such as conductance电导, potential电位, current电流, or quantity of electricity电量.



Features of electrochemical analysis:

(1) High accuracy, good reproducibility and stability

(2) High sensitivity

(3) Good selectivity (elimination of interference)

(4) Wide application (major, minor, and trace analysis)

(5) Simple instruments and easy automation

（1）准确度高，重现性和稳定性好

（2）灵敏度高

（3）选择性好（排除干扰）

（4）应用广泛（常量、微量和痕量分析）

（5）仪器设备简单，易于实现自动化

Section 1 Overview

Classification of electrochemical analysis:

1. Potentiometry: direct potentiometry, potentiometric titration

2. Electrolysis: electrogravimetry, coulometry, coulometric titration

3. Conductometry: direct conductometry, conductometric titration

4. Voltammetry: polarography, stripping methods, amperometric titration (dead-stop titration)

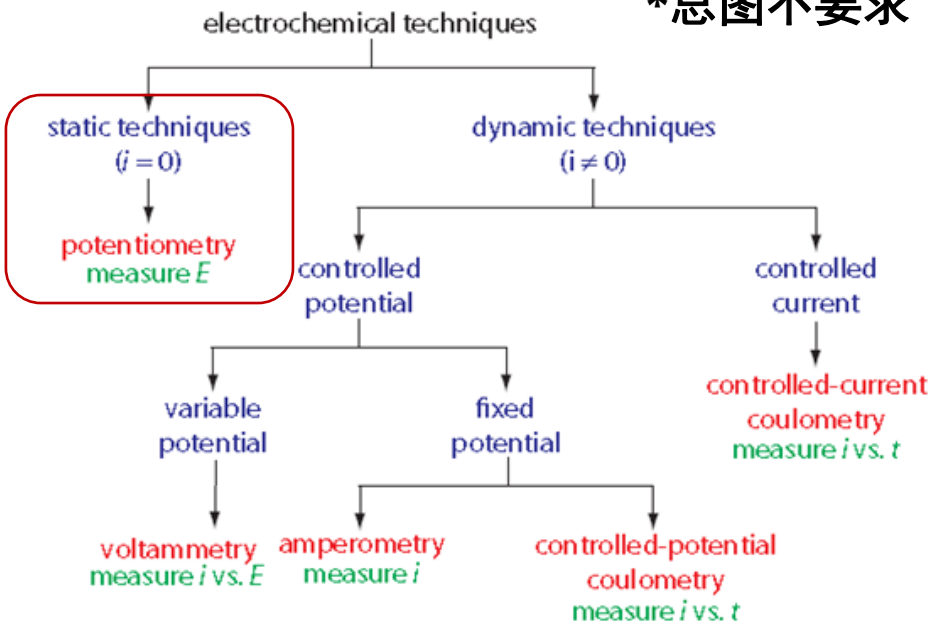
1.电位法　直接电位法、电位滴定法
2.电解法　电重量法、库仑法、库仑滴定法
3.电导法　直接电导法、电导滴定法
4.伏安法　极谱法、溶出法、电流滴定法（永停滴定法）

Section 1 Overview

Classification of electrochemical analysis:

- Movement of electrons in an oxidation–reduction reaction
- Measured in an electrochemical cell
- Potential, E
- Current, i

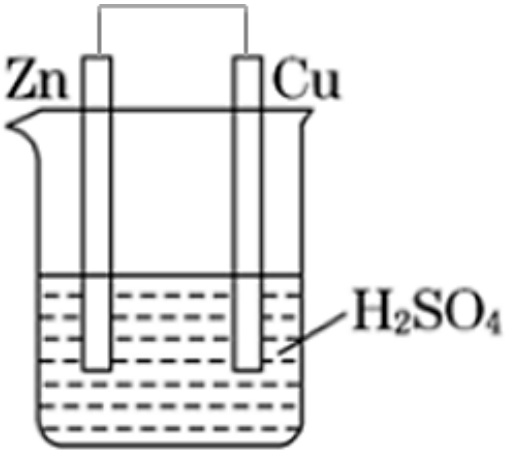
*总图不要求



Section 2 Basic Principles of Potentiometry

I. Chemical cells

A chemical cell化学电池is a device that realizes the interconversion between chemical reactions and electrical energy. It consists of two electrodes, an electrolyte solution, and an external circuit. It can be divided into galvanic cells原电池and electrolytic cells电解池.

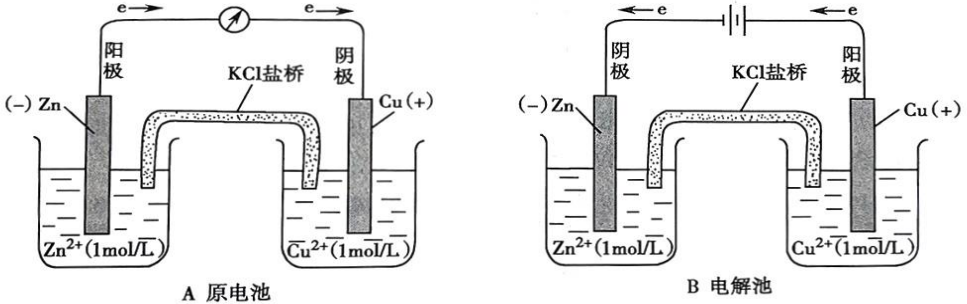


I. Chemical cells

(I) Galvanic cells and electrolytic cells原电池和电解池

A galvanic cell converts chemical energy into electrical energy, and its electrode reactions occur spontaneously电极反应自发进行.

An electrolytic cell converts electrical energy into chemical energy, and its electrode reactions occur only when an external voltage is applied.只有在外加电压的情况下，电极反应才能进行



I. Chemical cells

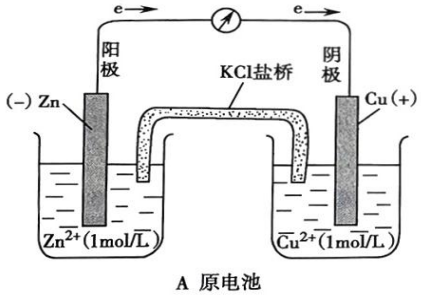
(I) Galvanic cells and electrolytic cells

Representation (Shorthand notation简写符号):

- Solutions are labeled with concentration or activity; gases should be labeled with pressure and temperature.溶液注明浓（活）度；气体应注明压力、温度
- Use | to represent each liquid-junction interface液接界面 in the cell.
- Use ¦ ¦ to represent two liquid-junction interfaces具有两个液接界面，|| to represent a salt bridge盐桥
- Write the electrode where oxidation occurs on the left and the electrode where reduction occurs on the right.发生氧化反应的电极写在左边，发生还原反应的电极写在右边

I. Chemical cells

由英国化学家 John Frederic Daniell 在 1836 年发明



A 原电池

(I) Galvanic cells and electrolytic cells

The Daniell galvanic cell is represented as: $Zn|ZnSO_4$ (1 mol/L)$||CuSO_4$ (1 mol/L)$|Cu$

The electrode half-reactions are:

Zinc electrode: $Zn \rightleftharpoons Zn^{2+} + 2e$ (oxidation, anode, negative氧化反应，阳极，负极)

Copper electrode: $Cu^{2+} + 2e \rightleftharpoons Cu$ (reduction, cathode, positive还原反应，阴极，正极)

Overall cell reaction: $Zn + Cu^{2+} \rightleftharpoons Zn^{2+} + Cu$

Under zero-current conditions, the electromotive force (EMF电动势) of the galvanic cell is:

$$E = \varphi_{(+)} - \varphi_{(-)} = \varphi^{\ominus}_{Cu^{2+}/Cu} - \varphi^{\ominus}_{Zn^{2+}/Zn} = (+0.337) - (-0.763) = 1.100\text{V}$$

I. Chemical cells

(I) Galvanic cells and electrolytic cells

Reverse voltage < 1.100 V; electrode reactions and current direction remain unchanged, but the current decreases. Galvanic cell

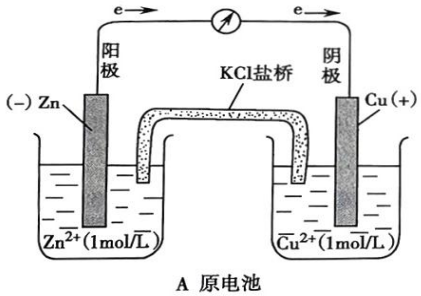
电极反应及电流方向不变，但电流变小->原电池

Reverse voltage = 1.100 V; no electrode reaction; current is zero.

无电极反应，电流为零

Reverse voltage > 1.100 V; electrode reactions and current direction both change.

Electrolytic cell电极反应及电流方向均发生改变->电解池



A 原电池

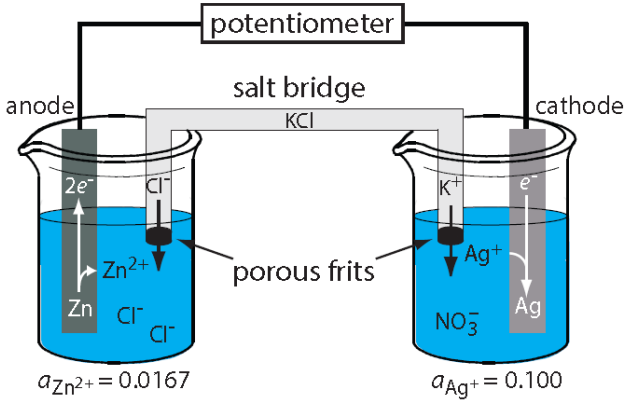
I. Chemical cells

由英国化学家 John Frederic Daniell 在 1836 年发明

(I) Galvanic cells and electrolytic cells

The Daniell galvanic cell is represented as: $Zn|ZnSO_4$ (1 mol/L)$||CuSO_4$ (1 mol/L)$|Cu$

Practice: What is the shorthand notation and what are the anodic and the cathodic reactions responsible for the potential of the electrochemical cell?



$$Zn\,(s) \rightleftharpoons Zn^{2+}\,(aq) + 2e^-$$

$$Ag^+\,(aq) + e^- \rightleftharpoons Ag\,(s)$$

$$Zn\,(s) + 2Ag^+\,(aq) \rightleftharpoons 2Ag\,(s) + Zn^{2+}\,(aq)$$

$$Zn\,(s)\,|\,ZnCl_2\,(aq, a_{Zn^{2+}} = 0.0167)\,\|\,AgNO_3\,(aq, a_{Ag^+} = 0.100)\,|\,Ag\,(s)$$

I. Chemical cells

(I) Galvanic cells and electrolytic cells

The electrolytic cell is represented as: $Cu|CuSO_4$ (1 mol/L)$||ZnSO_4$ (1 mol/L)$|Zn$

The electrode half-reactions are:

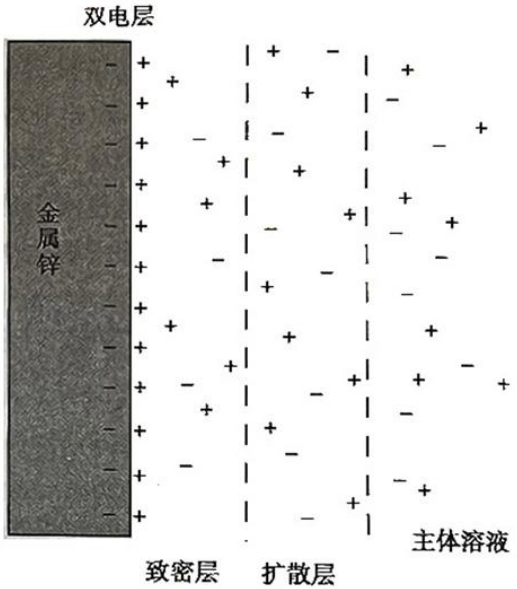
Zinc electrode: $Zn^{2+} + 2e \rightleftharpoons Zn$ (reduction, cathode还原反应、阴极)

Copper electrode: $Cu \rightleftharpoons Cu^{2+} + 2e$ (oxidation, anode氧化反应、阳极)

Overall cell reaction: $Zn^{2+} + Cu \rightleftharpoons Zn + Cu^{2+}$

I. Chemical cells

(II) Phase-boundary potential相界电位



Schematic diagram of the electric double layer

1. Electric double layer双电层

Formed at the interface between the metal and solution due to migration of charged particles在金属与溶液两相界面上，由于带电质点的迁移而形成

2. Phase-boundary potential or metal electrode potential相界电位或金属电极电位

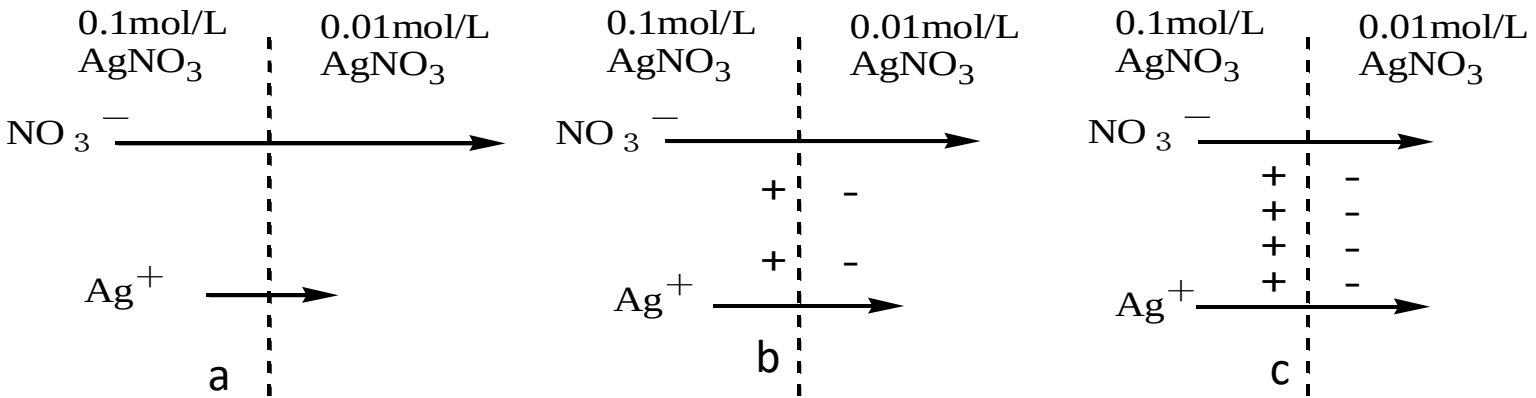
A stable electric double layer forms at the metal-solution interface and produces a potential difference.金属与溶液界面上形成了稳定的双电层而产生电位差

I. Chemical cells

Salt bridges can be used for elimination of liquid-junction potential可使用盐桥消除

(III) Liquid-junction potential液接电位

The potential difference produced at the contact interface between two electrolyte solutions of different composition, or of the same composition but different concentrations.在组成不同或组成相同而浓度不同的两个电解质溶液接触界面所产生的电位差



Schematic diagram of liquid-junction potential generation

I. Chemical cells

(III) Liquid-junction potential

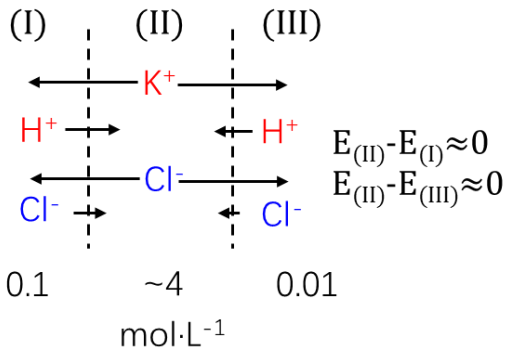
Elimination of liquid-junction potential: salt bridge盐桥

Composition and characteristics of a salt bridge:

- concentrated electrolyte solution高浓度电解质溶液
- Similar migration velocities of positive and negative ions 正负离子迁移速度相近

Functions of a salt bridge:

- prevent mixing of the two electrolyte solutions and reduce/eliminate the liquid-junction potential防止两种电解质溶液混和，降低/消除液接电位
- Provide an ion migration path (transfer electrons)提供离子迁移通道（传递电子）



II. Indicator and reference electrodes指示电极和参比电极

(I) Indicator electrode指示电极

An indicator electrode is an electrode whose potential changes with the activity (concentration) of the analyte (φ is related to c).

(II) Reference electrode参比电极

A reference electrode is an electrode whose potential remains essentially constant and does not change with the composition and concentration of the solution (φ is independent of c).

II. Indicator and reference electrodes

(I) Indicator electrode

Requirements for an indicator electrode in potentiometry: (1) its relation to the activity (concentration) of the analyte follows the Nernst equation; (2) fast response to the analyte, good reversibility, and good reproducibility; (3) simple structure, stable properties, and convenient use.

Common indicator electrodes
- Metal-based electrode金属基电极
- Membrane electrode膜电极

II. Indicator and reference electrodes

(I) Indicator electrode

1. Metal-based electrode: an electrode with a metal as the substrate and based on electron-transfer reactions是以金属为基体、基于电子转移反应的一类电极

(1) First-class electrode (metal-metal ion electrode):

Example: $Ag|Ag^+$

Electrode reaction: $Ag^+ + e \rightarrow Ag$

Electrode potential (25℃): $\varphi = \varphi^{\ominus} + 0.059\lg a_{Ag^+}$ *or* $\varphi = \varphi^{\ominus'} + 0.059\lg c_{Ag^+}$

Application: determination of the activity (concentration) of metal ions

II. Indicator and reference electrodes

1. Metal-based electrode

(2) Second-class electrode (metal-insoluble metal salt electrode):

Example: Ag|AgCl(s)|Cl^-

Electrode reaction: $AgCl + e \rightarrow Ag + Cl^-$

Electrode potential $\varphi = \varphi^{\ominus} + 0.059 \lg a_{Ag^+} = \varphi^{\ominus}_{Ag^+/Ag} + 0.059 \lg \frac{K_{SP,AgCl}}{a_{Cl^-}}$

$$= \varphi^{\ominus}_{Ag^+/Ag} + 0.059 \lg K_{SP,AgCl} - 0.059 \lg a_{Cl^-}$$

$$\varphi = \varphi^{\ominus}_{AgCl/Ag} - 0.059 \lg a_{Cl^-} \text{ or } \varphi = \varphi^{\ominus'}_{AgCl/Ag} - 0.059 \lg c_{Cl^-}$$

Application: determination of the concentration of the anion of a insoluble salt

II. Indicator and reference electrodes

(I) Indicator electrode

1. Metal-based electrode

(3) Zeroth-class electrode (inert metal electrode):

Example: Pt|Fe^{3+} ($a_{Fe}{}^{3+}$), Fe^{2+} ($a_{Fe}{}^{2+}$)

Electrode reaction: $Fe^{3+} + e \rightarrow Fe^{2+}$

Electrode potential (25 °C):$\varphi = \varphi^{\ominus} + 0.059\lg\frac{a_{Fe^{3+}}}{a_{Fe^{2+}}}$ 或$\varphi = \varphi^{\ominus\prime} + 0.059\lg\frac{c_{Fe^{3+}}}{c_{Fe^{2+}}}$.

Application: determination of the activity (concentration) or ratio of the oxidized and reduced forms测定氧化态、还原态活（浓）度或比值

II. Indicator and reference electrodes

$$E_{\text{cell}} = K + \frac{0.0592}{z}\lg(a_A)_{\text{samp}}$$

(I) Indicator electrode

2. Membrane electrode (ion-selective electrode)膜电极（离子选择电极）

(1) Primary electrode
原电极
基本电极

- Crystal electrode晶体电极 $\varphi = K - \frac{RT}{F}\ln\alpha_{F^-}$ 氟离子选择性电极
- Non-crystal electrode 非晶体电极
  - Rigid-matrix/glass electrode 刚性基质电极/玻璃电极
  - Mobile-carrier electrode 流动载体电极 $\varphi = K + \frac{RT}{2F}\ln\alpha_{Ca^{2+}}$ 钙离子选择性电极

$\varphi = K + \frac{RT}{nF}\ln\alpha_{M^{n+}}$ | $\varphi = K - \frac{RT}{nF}\ln\alpha_{M^{n-}}$

II. Indicator and reference electrodes

(I) Indicator electrode

2. Membrane electrode (ion-selective electrode)

(2) Gas-sensing electrode: hydrophobic, allowing only gas to pass through and not ions in solution气敏电极：具有疏水性，只允许气体通过，不允许溶液中离子通过

Example: carbon dioxide gas-sensing electrode. When carbon dioxide gas passes through the gas-permeable membrane into the internal solution (0.01 mol/L sodium bicarbonate electrolyte solution), the pH of the solution changes. The carbon dioxide content can be indirectly determined from the potential change of the pH glass electrode.
例：二氧化碳气敏电极，当二氧化碳气体透过透气膜进入中介液（0.01mol/L碳酸氢钠电解质溶液）时，则溶液的pH就会发生改变，通过pH玻璃电极的电位变化就可以间接测定二氧化碳的含量

II. Indicator and reference electrodes

(I) Indicator electrode

2. Membrane electrode (ion-selective electrode)

(3) Enzyme electrode:酶电极　　葡萄糖氧化酶glucose oxidase,GOD

$$葡萄糖 + O_2 + H_2O \xrightarrow{GOD} 葡萄糖酸 + H_2O_2$$

葡萄糖酶电极

阳极反应：$H_2O_2 \longrightarrow 2H^+ + O_2 + 2e$ （$\varphi^{\ominus} = +0.69V$）

阴极反应：$2H^+ + 2e + 1/2O_2 \longrightarrow H_2O$ （$\varphi^{\ominus} = +0.401V$）

总电池反应：$H_2O_2 \longrightarrow H_2O + 1/2O_2$

Electrochemical biosensor to determine O_2、Hydrogen peroxide H_2O_2 or glucose

II. Indicator and reference electrodes

(II) Reference electrode参比电极

An electrode whose potential remains essentially constant and does not change with the composition and concentration of the solution (φ is independent of c).

A reference electrode must meet the following conditions:

(1) constant electrode potential and good reversibility;

(2) good reproducibility and stability, and long service life;

(3) high impedance, low current density, and small temperature effect.

①电极电位保持不变，可逆性好；②重现性和稳定性好，使用寿命长；③阻抗大，电流密度小，温度影响小

II. Indicator and reference electrodes

(II) Reference electrode

1. Saturated calomel electrode (SCE)饱和甘汞电极

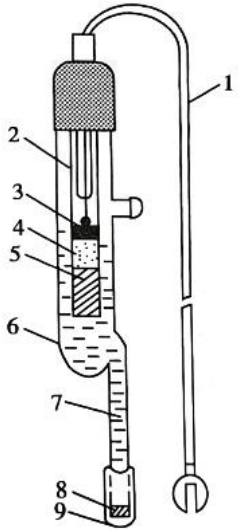
Represented as $Hg|Hg_2Cl_2(s)|KCl$

Electrode reaction: $Hg_2Cl_2 + 2e \rightarrow 2Hg + 2Cl^-$

Electrode potential: (25℃)： $\varphi = \varphi^{\ominus} - 0.059\lg a_{Cl^-}$ or $\varphi = \varphi^{\ominus'} - 0.059\lg c_{Cl^-}$

$$\varphi_{\text{SCE（饱和）}} = 0.2412\text{V}$$

Saturated calomel electrode



1.电极引线；2.玻璃内管；3.汞；4.汞-甘汞糊（Hg_2Cl_2和Hg研磨的糊）；5.石棉或纸浆；6.玻璃外管套；7.饱和KCl溶液；8.素烧瓷片；9.小橡皮塞。

II. Indicator and reference electrodes

(II) Reference electrode

2. Silver-silver chloride electrode银-氯化银电极

Represented as Ag|AgCl|Cl^-

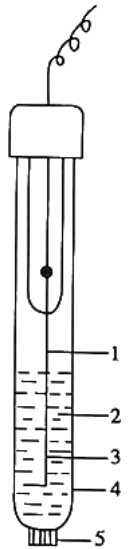
Electrode reaction: AgCl + e → Ag + Cl^-

Electrode potential (25 °C):$\varphi = \varphi^{\ominus} - 0.059\lg a_{Cl^-}$

$$\text{or } \varphi = \varphi^{\ominus'} - 0.059\lg c_{Cl^-}$$

Saturated KCl solution $\varphi = 0.1990V$

Silver-silver chloride electrode



1.银丝；2.饱和KCl溶液；3.银-氯化银；4.玻璃管；5.素烧瓷芯。

Section 3 Direct Potentiometry

Section 3 Direct Potentiometry

Direct potentiometry直接电位法: a method in which a suitable indicator electrode and reference electrode are selected, immersed in the test solution to form a galvanic cell, the electromotive force of the galvanic cell is measured, and the activity (concentration) of the analyte in the test solution is obtained from the Nernst equation.选择合适的指示电极与参比电极，浸入待测溶液中组成原电池，测量原电池的电动势，根据能斯方程求得待测溶液中被测组分活（浓）度的方法

I. Measurement of solution pH

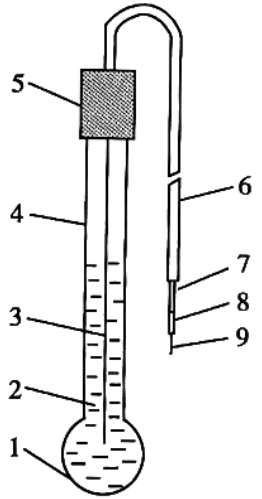
(I) pH glass electrode pH 玻璃电极

1. Structure and response mechanism of the pH glass electrode

Glass electrode membrane: contains 21.4% Na_2O, 6.4% CaO, and 72.2% SiO_2, Thickness about 0.05-0.1 mm; selective response to H^+

Internal reference solution: KCl buffer solution at pH 7 or pH 4

Internal reference electrode: Ag-AgCl electrode



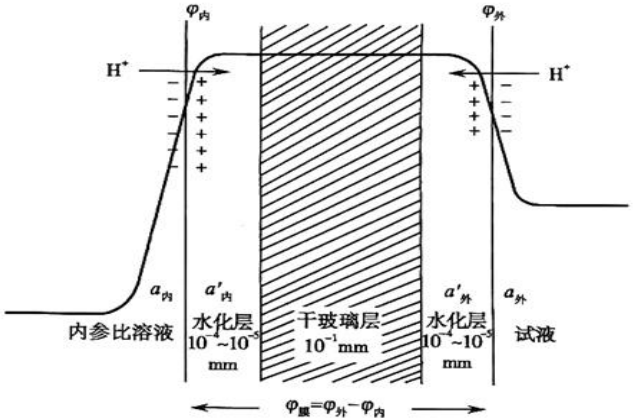
pH glass electrode

1.玻璃膜球；2.内参比溶液；3.Ag-AgCl电极；4.玻璃管；5.电极帽；6.外套管；7.网状金属屏；8.高绝缘塑料；9.电极导线。

I. Measurement of solution pH

(I) pH glass electrode

1. Structure and response mechanism of the pH glass electrode



Schematic diagram of glass electrode membrane layers and potential generation

- After sufficient soaking in water

-> H^+ can penetrate into the glass membrane

-> H^+ exchanges with Na^+

-> Hydrated gel layer水化凝胶层/dry glass layer干玻璃层

- Immersed in the test solution

-> H^+ diffusion扩散

-> Electric double layer forms双电层, producing a potential difference电位差

-> Stable phase-boundary potential is reached (outer/inner)

Phase-boundary potential generation: result of ion exchange and diffusion相界电位产生——离子交换和扩散的结果

I. Measurement of solution pH

1. Structure and response mechanism of the pH glass electrode

Phase-boundary potential between the external solution and external gel layer

$$\varphi_{外} = K_1 + \frac{2.303RT}{F}\lg\frac{a_{外}}{a'_{外}}$$

Phase-boundary potential between the internal solution and internal gel layer

$$\varphi_{内} = K_2 + \frac{2.303RT}{F}\lg\frac{a_{内}}{a'_{内}}$$

When $K_1 = K_2$, $a'_{内} = a'_{外}$, $a_{内}$ is constant:

$$膜电位\ \phi_m = \phi_{外} - \phi_{内} = \frac{2.303RT}{F}\lg\frac{a_{外}}{a_{内}} = K' + \frac{2.303RT}{F}\lg a_{外}$$

I. Measurement of solution pH

1. Structure and response mechanism of the pH glass electrode

For the whole glass electrode, its electrode potential is:

$$\varphi_{玻璃} = \varphi_{内参比} + \varphi_{膜} = \varphi_{AgCl/Ag^+} + \left(K' + \frac{2.303RT}{F}\lg a_{外}\right).$$

$$= \left(\varphi_{AgCl/Ag} + K'\right) - \frac{2.303RT}{F}pH = K - \frac{2.303RT}{F}pH$$

$$\varphi = K - 0.059pH$$

The glass electrode potential is linearly related to the pH of the test solution. This is the theoretical basis for using the pH glass electrode as an indicator electrode to measure solution pH.玻璃电极的电位φ与待测试液的pH呈线性关系，这是pH玻璃电极作为指示电极测定溶液pH的理论依据

I. Measurement of solution pH

2. Performance of the pH glass electrodepH玻璃电极的性能

(1) Conversion coefficient转换系数: the change in glass electrode potential caused by a change of one pH unit每改变一个pH单位引起玻璃电极电位的变化值

$$S = \frac{2.303RT}{F}$$

$$S = -\frac{\Delta\varphi}{\Delta pH}$$

$theoretically\ S$=59mV（25℃）

I. Measurement of solution pH

2. Performance of the pH glass electrode

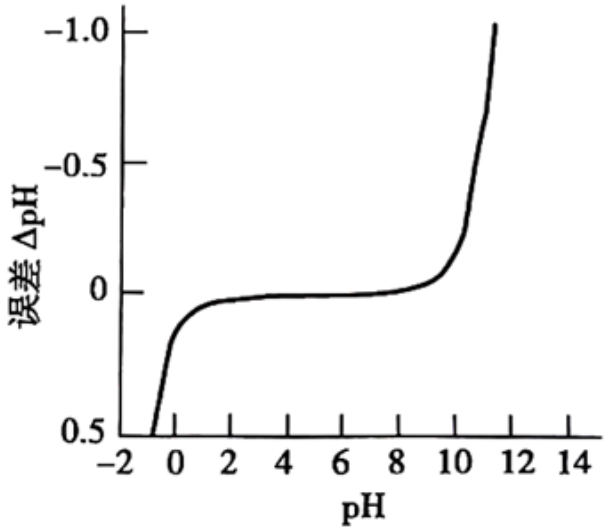
(2) Alkaline error and acid error:碱差与酸差

- generally, only at pH 1-9 does the φ-pH curve of a glass electrode show good linearity.

Alkaline error or sodium error:碱差或钠差

- pH > 9, measured pH < true pH → negative error

Acid error: pH < 1, measured pH > true pH → positive error



Acid and alkaline errors of the glass electrode

I. Measurement of solution pH

2. Performance of the pH glass electrode

(3) Asymmetry potential:不对称电位 $\varphi_{膜} = \frac{2.303RT}{F} \lg \frac{a_{外}}{a_{内}}$

When a(outer) = a(inner) (the pH inside and outside the membrane is the same), it is not 0.

- Causes: the structure and properties of the inner and outer surfaces of the glass membrane are not exactly the same, contamination of the outer surface, chemical corrosion, mechanical damage, and other factors.玻璃膜内外表面的结构和性能不完全相同、外表面玷污、化学腐蚀和机械损伤等因素所致
- Before use, the glass electrode must be fully soaked in water for a certain period. This activates the electrode活化电极and also minimizes and stabilizes the asymmetry potential.

I. Measurement of solution pH

2. Performance of the pH glass electrode

(4) Internal resistance of the electrode电极的内阻: very large; only a very small current is allowed to pass很大，只允许有微小的电流通过

(5) Working temperature: 0-50 °C

Application characteristics

Advantages: direct measurement is convenient and does not destroy the solution; suitable for pH measurement of colored and turbid liquids直接测量方便，不破坏溶液，适于有色、浑浊液体的pH的测定

Disadvantages: thin glass membrane; easily damaged玻璃膜薄，易损

I. Measurement of solution pH

(II) Method for measuring solution pH

In direct potentiometry for solution pH, a pH glass electrode is commonly used as indicator electrode and a saturated calomel electrode as reference electrode; they are immersed in the test solution to form a galvanic cell:直接电位法测定溶液pH，常用pH玻璃电极作为指示电极，饱和甘汞电极作为参比电极，浸入待测液中组成原电池

(-) Ag|AgCl(s), internal filling solution|glass membrane|test solution||KCl(saturated), Hg_2Cl_2(s)|Hg (+)

The EMF of the galvanic cell is: E =$\varphi_{(SCE)}$ - $\varphi_{(glass)}$

$$E = \varphi_{SCE} - \left(K - \frac{2.303RT}{F}\text{pH}\right)$$

Under certain conditions, it is constant, therefore: $E = K' + \frac{2.303RT}{F}\text{pH}$

I. Measurement of solution pH

(II) Method for measuring solution pH

Under the same conditions, first measure the EMF of a standard buffer solution of known pH:在相同的条件下，首先测量已知pH标准缓冲溶液的电动势

$$E_s = K' + \frac{2.303RT}{F}\,\mathrm{pH_s}$$

Then measure the EMF of the test solution然后测量待测溶液电动势:

$$E_x = K' + \frac{2.303RT}{F}\,\mathrm{pH_x}$$

Subtracting the two equations gives: $\mathrm{pH_x} = \mathrm{pH_s} + \frac{E_x - E_s}{2.303RT/F}$

I. Measurement of solution pH

(II) Method for measuring solution pH

Precautions for using a pH glass electrode to measure solution pH: (1) an ordinary pH glass electrode is suitable for pH 1-9; (2) pHs of the standard buffer should be as close as possible to pHx of the test solution; (3) the glass electrode should be soaked in distilled water for more than 24 h before use; 使用pH玻璃电极测量溶液pH应注意：①普通pH玻璃电极适用测量范围pH 为1～9；②标准缓冲溶液的pH_s应尽量与待测液的pH_x接近；③玻璃电极使用前需在蒸馏水中浸泡24小时以上；

I. Measurement of solution pH

(II) Method for measuring solution pH

Precautions for using a pH glass electrode to measure solution pH: (4) the temperatures of the standard buffer and test solution must be the same and kept as constant as possible; (5) preparation, use, and storage of the standard buffer must strictly follow specifications; (6) because of the glass membrane, the glass electrode cannot be used for pH measurement of acidic fluoride-containing solutions. ④标准缓冲溶液与待测液的温度必须相同并尽量保持恒定；⑤标准缓冲溶液的配制、使用、保存应严格按规定进行；⑥由于F^-腐蚀玻璃膜，因此，玻璃电极不能用于含氟化物酸性溶液的pH测定

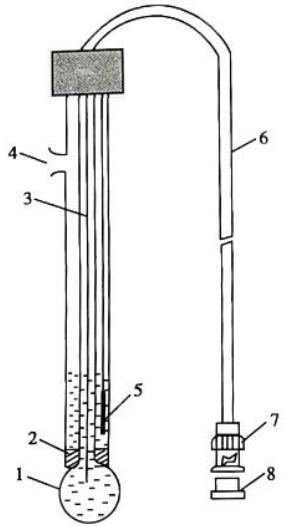
I. Measurement of solution pH

(III) Composite pH electrode复合pH电极

A glass electrode and a reference electrode (calomel electrode or silver-silver chloride electrode) are combined into a single electrode body.

Features:

- Small size体积小
- Convenient, sturdy, and durable使用方便、坚固、耐用
- Suitable for pH measurement of small-volume solutions有利于小体积溶液pH测定



Composite pH electrode

1.玻璃电极；2.瓷塞；3.内参比电极；4.充液口；5.参比电极体系；6.导线；7.插口；8.防尘塞。

II. Measurement of other ion concentrations

(I) Ion-selective electrodes

1. Structure and response mechanism of ion-selective electrodes

When the electrode is immersed in a solution containing the response ion, an electric double layer forms at the interface between the electrode membrane and solution, producing a stable membrane potential.当电极浸入响应离子溶液后，在电极膜和溶液界面间形成双电层，产生稳定的膜电位

$$\varphi = K \pm \frac{2.303RT}{nF} lg\, \alpha_i = K' \pm \frac{2.303RT}{nF} lg\, c_i$$

$K \rightarrow$ activity electrode constant, $K' \rightarrow$ concentration electrode constant, n denotes the charge number of the responding ion; positive sign for cations and negative sign for anions

K→活度电极常数，K'→浓度电极常数，n为响应离子的电荷数，阳离子取正号，阴离子取负号

II. Measurement of other ion concentrations

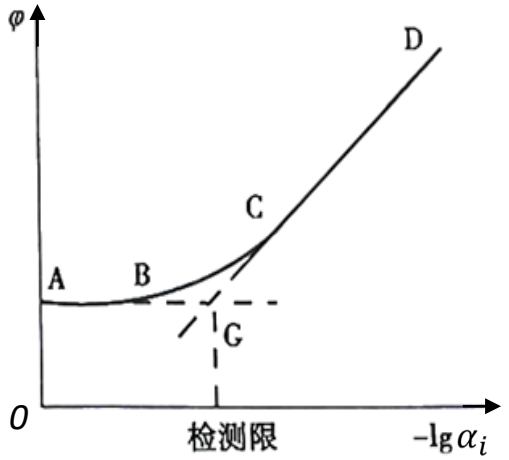
(I) Ion-selective electrodes

2. Performance of ion electrodes

(1) Linear range: the range of ion activity (concentration) corresponding to the linear portion of the working curve is called the linear range of an ion-selective electrode.

(2) Detection limit: the lowest concentration of the target ion that an ion-selective electrode can detect

把线性段 CD 和低浓度平坦段 BA 分别延长，交点 G 对应的横坐标活度，就是IUPAC定义的检测下限，代表该电极可以可靠定量的最低活度。



Working curve and detection limit of ion-selective electrodes

II. Measurement of other ion concentrations

(I) Ion-selective electrodes

2. Performance of ion electrodes

(3) Selectivity coefficient选择性系数: under the same conditions, the ratio of the response ability of the same electrode to ion Y (interfering ion) and ion X (target ion), that is, the activity ratio of X and Y that provides the same potential response在相同条件下，同一电极对Y（干扰离子）和X（被测离子）离子响应能力之比，亦即提供相同电位响应的X离子和Y离子的活度比:

$$K_{X,Y} = \frac{\alpha_X}{(\alpha_Y)^{n_X/n_Y}}$$

The smaller the value of $K_{X,Y}$, the higher the selectivity of the electrode for ion X and the smaller the interference from ion Y.

Example: glass electrode $K_{H^+,\,Na^+} = 10^{-11}$; this indicates that the response of the electrode to H^+ is 10^{11} times higher than its response to Na^+.

II. Measurement of other ion concentrations

(I) Ion-selective electrodes

2. Performance of ion electrodes

(4) Effective pH range: refers to the applicable pH range of an ion-selective electrode. Exceeding this range will result in significant errors.有效pH范围：指离子选择电极的pH使用范围，超出该范围会产生较大误差。

(5) Response time: the time required for the battery electromotive force to become stable. Stirring can be used to increase the diffusion rate and shorten this time.响应时间：电池电动势达到稳定所需的时间。可通过搅拌提高扩散速率，缩短时间。

II. Measurement of other ion concentrations

(II) Methods for measuring ion concentration

An ion-selective electrode for the target ion is used as indicator electrode and a saturated calomel electrode (SCE) as reference electrode; they are immersed in the test solution to form a galvanic cell:

(-) Ion-selective electrode | test solution || KCl(saturated), $Hg_2Cl_2(s)|Hg$ (+)

The EMF of the cell is: $E = \varphi(SCE) - \varphi(ISE)$

$$E = \varphi_{SCE} - \left(K' \pm \frac{2.303RT}{nF} \lg c_i \right) = K \mp \frac{2.303RT}{nF} \lg c_i$$

II. Measurement of other ion concentrations

(II) Methods for measuring ion concentration

During measurement, add an equal amount of TISAB (total ionic strength adjustment buffer总离子强度调节剂)

TISAB: a concentrated electrolyte solution that does not contain the target ion, does not react with the target ion, and does not contaminate or damage the electrode membrane一种不含被测离子、不与被测离子反应、不污染或损害电极膜的浓电解质溶液

II. Measurement of other ion concentrations

(II) Methods for measuring ion concentration

1. Direct comparison method (two-measurement method)直接比较法（两次测量法）

$$E_x - E_s = \mp \frac{2.303RT}{nF} lg（c_x - c_s）$$

Note: (1) use "-" for cations and "+" for anions; (2) the standard solution concentration should be close to the target ion concentration

①阳离子取“－”，测阴离子取“+”；②标准溶液浓度应与待测离子浓度相近

II. Measurement of other ion concentrations

(II) Methods for measuring ion concentration

2. Working curve method工作曲线法:

prepare a standard series of solutions using a reference standard of the target ion (matrix should be the same as the test solution), measure the EMF of each with the selected indicator and reference electrodes from low to high concentration, and plot Es versus lg Cs.用待测离子的对照品配制标准系列溶液（基质应与试液相同），选用的指示电极和参比电极按浓度从低到高分别测量其电动势，绘制E_s对lgC_s曲线

II. Measurement of other ion concentrations

(II) Methods for measuring ion concentration

2. Working curve method工作曲线法:

Advantages: usable even when the electrode response does not completely obey the Nernst equation.即使电极响应不完全服从Nernst方程的也可用

Requirements: standard solution composition similar to test solution composition and same solution temperature标液组成与试液组成相近，溶液温度相同

Standard solution and test solution have the same ionic strength and the same activity coefficient (add an equal amount of TISAB)标液与试液离子强度一致，活度系数相同（等量加入TISAB）

$$E_{cell} = K + \frac{0.059}{z_A} \lg[A]$$

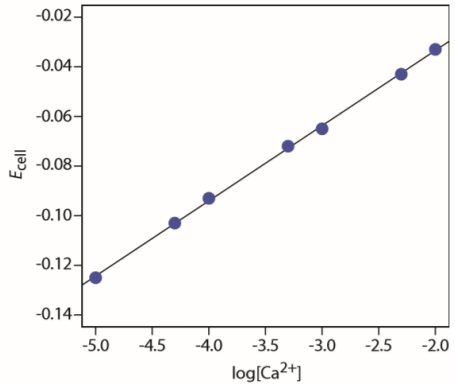
2. Working curve method工作曲线法:Practice

The concentration of Ca^{2+} in a water sample is determined using the method of external standards. The ionic strength of the samples and the standards is maintained at a nearly constant level by making each solution 0.5 M in KNO_3. The measured cell potentials for the external standards are shown in the following table.

$[Ca^{2+}]$ (M)	E_{cell} (V)
1.00×10^{-5}	–0.125
5.00×10^{-5}	–0.103
1.00×10^{-4}	–0.093
5.00×10^{-4}	–0.072
1.00×10^{-3}	–0.065
5.00×10^{-3}	–0.043
1.00×10^{-2}	–0.033

What is the concentration of Ca^{2+} in a water sample if its cell potential is found to be –0.084 V?

- $E_{cell} = K + \frac{0.059}{2} \lg[Ca^{2+}]$
- Linear regression
- $E_{cell} = 0.027 + 0.0303 \lg[Ca^{2+}]$
- When E_{cell} = -0.084 V
- $[Ca^{2+}] = 2.17 \times 10^{-4}$



II. Measurement of other ion concentrations

(II) Methods for measuring ion concentration

(3) Standard addition method:标准加入法

Measure the EMF E_X of a test solution with volume V_X and concentration C_X, then add a standard solution of the target ion with concentration C_S ($C_S > 10C_X$) and volume V_S ($V_S < V_X/10$) to the test solution, and measure the EMF E;

测定体积V_X、浓度为C_X的待测试液电动势E_X，然后向试液中加入浓度为C_S（C_S>10C_X），体积V_S（V_S< V_X/10），的待测离子标准溶液，测得电动势为E

$$E_x = K \mp \frac{2.303RT}{nF} \lg c_x, \quad E = K \mp \frac{2.303RT}{nF} \lg \frac{c_x V_x \times c_s V_s}{V_x + V_s}$$

II. Measurement of other ion concentrations

(II) Methods for measuring ion concentration

(3) Standard addition method:

let $S=\mp\frac{2.303RT}{nF}$,then $\triangle E = E - E_x = S\lg\frac{c_x V_x \times c_s V_s}{V_x + V_s}$:

$$C_x = \frac{c_s V_s}{V_x + V_s}(10^{\triangle E/S} - \frac{V_x}{V_x + V_s})^{-1}$$

Since $V_x \gg V_s$, it can be considered $V_x + V_s \approx V_x$, the above equation can be simplified to: $C_x = \frac{c_s V_s}{V_x}(10^{\triangle E/S} - 1)^{-1}$

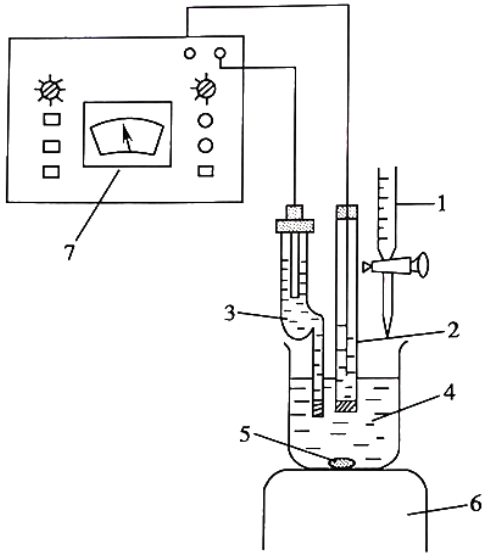
Advantages: no need to add TISAB; simple and rapid operation

Section 4 Potentiometric Titration

I. Basic Principles

Potentiometric titration电位滴定法

- A type of titrimetric analysis method in which the titration endpoint is determined from the change in cell EMF during titration.
- It is suitable for titrations in which no suitable indicator is available, the solution is deeply colored or turbid, and the endpoint is difficult to judge with an indicator. It has high accuracy and is easy to perform continuously, automatically, and on a micro scale.
- 适用于没有合适指示剂、溶液颜色较深或混浊，难于用指示剂判断终点的滴定分析法，准确度高、易实现连续、自动和微量滴定



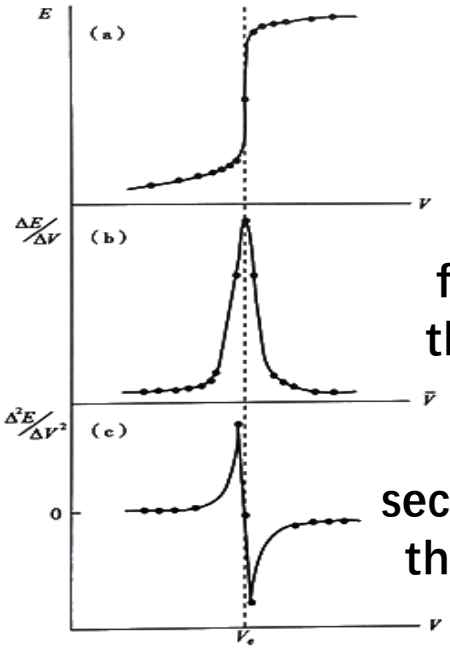
Schematic diagram of potentiometric titration apparatus

1.滴定管；2.指示电极；3.参比电极；4.待测溶液；5.搅拌子；6.电磁搅拌器；7.电位测量仪。

I. Basic Principles

Methods for determining the titration endpoint:

1. E-V curve method
2. $\Delta E/\Delta V$-$\bar{V}$ curve method
3. $\Delta^2 E/\Delta V^2$ -V curve method



titration curve

first derivative of the titration curve

second derivative of the titration curve

Endpoint Determination

(a) E-V; (b) $\Delta E/\Delta V$- $\bar{V}$; (c) $\Delta^2 E/\Delta V^2$ -V .

I. Basic Principles

Draw the derivative of the titration curves

titration curve

Volume (mL)	pH
23.65	6.00
23.91	6.10
24.13	6.20

first derivative of the titration curve

Volume (mL)	ΔpH/ΔV
23.78	0.385
24.02	0.455

second derivative of the titration curve

Volume (mL)	Δ^2pH/ΔV^2
23.90	0.292

$$\frac{\Delta \mathrm{pH}}{\Delta \mathrm{V}} = \frac{6.10 - 6.00}{23.91 - 23.65} = 0.385$$

$$\frac{\Delta^2 \mathrm{pH}}{\Delta \mathrm{V}^2} = \frac{0.455 - 0.385}{24.02 - 23.78} = 0.292$$

I. Basic Principles

Draw the derivative of the titration curves - practice

titration curve

Volume (mL)	pH
22.15	7.00
22.41	7.20
22.63	7.40

first derivative of the titration curve

Volume (mL)	$\Delta pH/\Delta V$

second derivative of the titration curve

Volume (mL)	$\Delta^2 pH/\Delta V^2$

I. Basic Principles

Draw the derivative of the titration curves - practice

titration curve

Volume (mL)	pH
22.15	7.00
22.41	7.20
22.63	7.40

first derivative of the titration curve

Volume (mL)	ΔpH/ΔV
22.28	0.790
22.52	0.910

second derivative of the titration curve

Volume (mL)	Δ^2pH/ΔV^2
22.40	0.584

$$\frac{\Delta \mathrm{pH}}{\Delta \mathrm{V}} = \frac{7.20 - 7.00}{22.41 - 22.15} = 0.790$$

$$\frac{\Delta^2 \mathrm{pH}}{\Delta \mathrm{V}^2} = \frac{0.910 - 0.790}{22.52 - 22.28} = 0.584$$

II. Applications of potentiometric titration

(I) Types of potentiometric titration

1. Acid-base titration

- Used to determine the equilibrium constant平衡常数of a weak acid or weak base
- For example, when NaOH titrates a monoprotic weak acid HA, at the half-neutralization point [HA] = [A^-], Ka = [H^+], that is, pKa = pH.

II. Applications of potentiometric titration

(I) Types of potentiometric titration

1. Acid-base titration

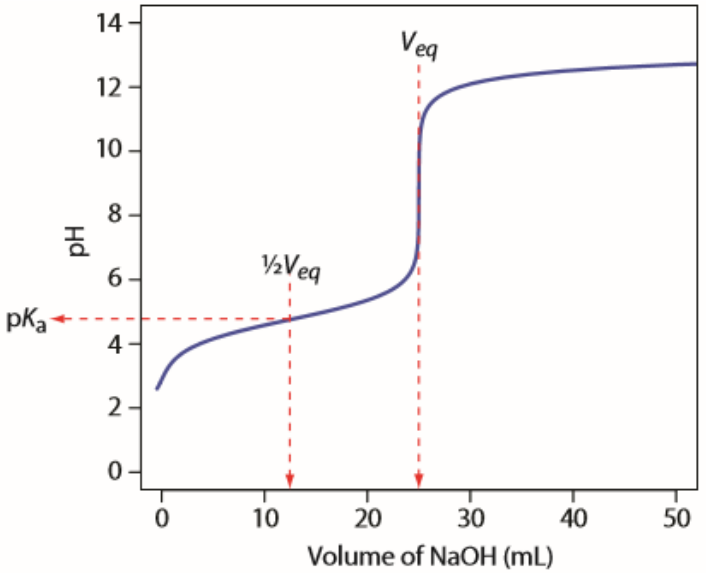
- Used to determine the equilibrium constant平衡常数of a weak acid or weak base

For example, Dissociation constant of CH_3COOH

$$K_a = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]}$$

When $[CH_3COOH] = [CH_3COO^-]$

The pH equals the pK_a when the volume of NaOH is approximately half of the V_{eq}



II. Applications of potentiometric titration

(I) Types of potentiometric titration

2. Precipitation titration

3. Complexometric titration

4. Redox titration

	指示电极	参比电极
➢ 酸碱滴定法：	pH玻璃电极	SCE
➢ 沉淀滴定法：	Ag 电极	SCE
➢ 配位滴定法：	铂电极	SCE
➢ 氧化还原滴定法：	铂电极	SCE

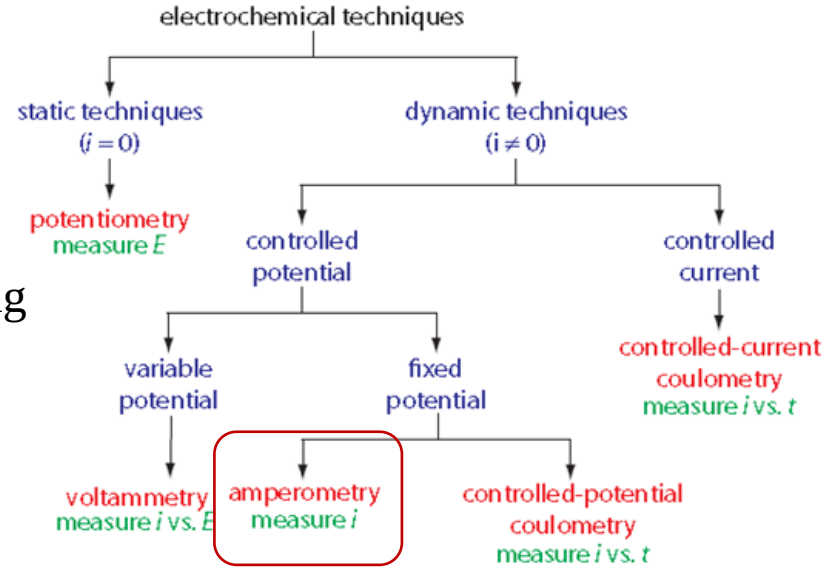
Section 5 Dead-Stop Titration

I. Basic principles of dead-stop titration

Dead-stop titration永停滴定法:

also called dual-current or dual-amperometric titration. It is a method for determining the titration endpoint from changes in current during titration and is a type of amperometric titration.

又称双电流或双安培滴定法。根据滴定过程中的电流变化来确定滴定终点的方法，属于电流滴定法的一种



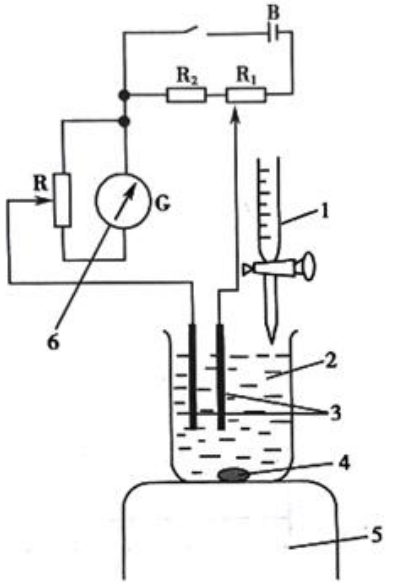
Section 5 Dead-Stop Titration

I. Basic principles of dead-stop titration

Measurement principle:

Insert two identical Pt electrodes into the sample solution, apply a low voltage between the two electrodes, connect an ammeter电流计, and perform titration. The endpoint is determined from the change in the ammeter needle.

将两个相同Pt电极插入样品溶液中，在两极之间外加低电压，连电流计，进行滴定，通过电流计指针的变化确定终点



Schematic diagram of dead-stop titration apparatus

1.滴定管；2.待测溶液；3.Pt电极；4.搅拌子；5.电磁搅拌器；6.电流测量仪。

I. Basic principles of dead-stop titration

Reversible redox couple: Both the oxidized state and reduced state of the redox couple can undergo rapid oxidation/reduction reactions on the platinum electrode, and the electrode reaction is fast and reversible. When Ox and Red coexist in the solution, an obvious electrolytic current will be generated.

可逆电对：电对的氧化态、还原态在铂电极上都能快速发生氧化/还原反应，电极反应迅速可逆。溶液同时存在Ox和Red 时，就会产生明显电解电流。

I. Basic principles of dead-stop titration

Reversible redox couple: halogens / halide ions, cerium, iron卤素 / 卤离子、铈、铁等

电对	电极反应	应用场景	永停曲线特点
I_2/I^-	$I_2 + 2e^- \rightleftharpoons 2I^-$	碘量法，硫代硫酸钠滴定碘、碘滴定硫代硫酸钠	半计量点电流最大；计量点附近电流降到接近 0
Fe^{3+}/Fe^{2+}	$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	硫酸铈滴定亚铁，重铬酸钾滴定亚铁	半计量点电流峰值，典型可逆电对电流变化
Ce^{4+}/Ce^{3+}	$Ce^{4+} + e^- \rightleftharpoons Ce^{3+}$	铈量法滴定还原剂	可逆，铂电极响应快
Br_2/Br^-	$Br_2 + 2e^- \rightleftharpoons 2Br^-$	溴量法	和碘电对行为类似

I. Basic principles of dead-stop titration

Irreversible redox couple: At least one of the oxidized state and reduced state has an extremely slow reaction rate on the platinum electrode and is not prone to electrode reaction; even if both coexist, almost no current is produced.

不可逆电对：氧化态、还原态中至少一种，在铂电极上反应速率极慢，不容易发生电极反应；即便两者共存，也几乎不产生电流。

I. Basic principles of dead-stop titration

Irreversible redox couple

电对	说明	应用场景	永停曲线特点
$HNO_2/Ar-NH_2$（亚硝酸 / 芳伯胺）	芳伯胺被氧化、亚硝酸被还原，**电极反应很慢**；滴定终点前，体系只有芳伯胺，没有足够HNO_2，几乎无电流；终点稍微过量一滴$NaNO_2$，溶液出现HNO_2电对，电流立刻持续上升	中国药典亚硝酸钠滴定，测磺胺类药物	终点前电流≈0；终点突跃，电流突然持续增大
MnO_4^-/Mn^{2+}	虽然氧化还原反应快，但在铂电极表面电极动力学慢，属于不可逆电对	高锰酸钾滴定，一般不用永停	不容易产生稳定电流
$S_4O_6^{2-}/S_2O_3^{2-}$	硫代硫酸根 / 连四硫酸根，电极反应不可逆	碘滴定硫代硫酸钠时，这个电对本身不可逆，体系里是I_2/I^-可逆电对贡献电流	电流来自碘电对，不是硫代 / 连四硫酸根

I. Basic principles of dead-stop titration

Determination Principle of the Reversible Redox Couple I_2/I^-

$$\varphi_{I_2/I^-} = \varphi^{\ominus\prime}_{I_2/I^-} + \frac{0.059}{2}\lg\frac{c_{I_2}}{c^2_{I^-}} \quad (25℃)$$

Electrolytic cell (I_2/I^-):

Anode (oxidation): $2I^- \rightleftharpoons I_2 + 2e$ 阳极（氧化反应）

Cathode (reduction): $I_2 + 2e \rightleftharpoons 2I^-$ 阴极（还原反应）

I. Basic principles of dead-stop titration

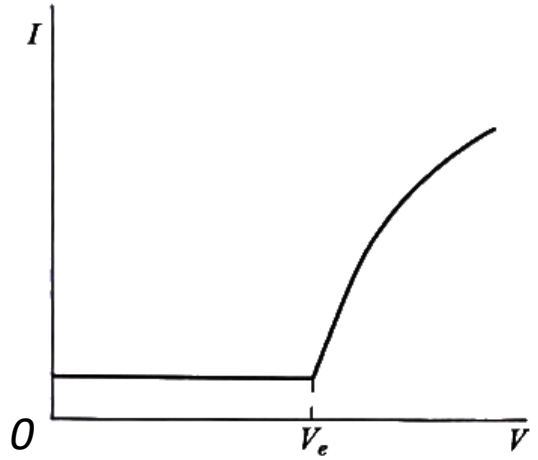
Features: during titration, the current is maximum when the concentrations of the oxidized and reduced forms of the reaction couple are equal. When they are unequal, the current is determined mainly by the concentration of the less abundant oxidized or reduced form.在滴定过程中，反应电对氧化态和还原态的浓度相等时，电流最大，若浓度不相等时，电流大小则由浓度小的氧化态或还原态的浓度决定（因总物质的量 [Ox]+[Red] = $C_{总}$（固定））

I. Basic principles of dead-stop titration

1. Reversible couple titrating irreversible couple可逆电对滴定不可逆电对

$$I_2 + 2S_2O_3^{2-} \rightleftharpoons 2I^- + S_4O_6^{2-}$$

Before the stoichiometric point, only I^- and the irreversible couple $S_2O_3^{2-}/S_4O_6^{2-}$ are present, and there is no current; after the stoichiometric point, current is produced.



Current-change curve for I_2 titration of $Na_2S_2O_3$

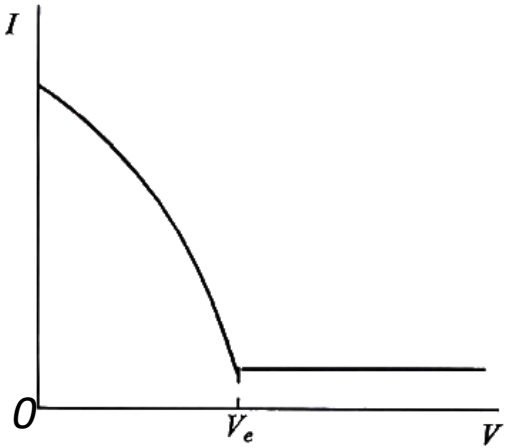
I. Basic principles of dead-stop titration

2. Irreversible couple titrating reversible couple不可逆电对滴定可逆电对

$$I_2 + 2S_2O_3^{2-} \rightleftharpoons 2I^- + S_4O_6^{2-}$$

$Na_2S_2O_3 \rightarrow I_2$

Current is present initially, and the current decreases to a minimum at the endpoint.



Current-change curve for $Na_2S_2O_3$ titration of I_2

I. Basic principles of dead-stop titration

3. Reversible couple titrating reversible couple可逆电对滴定可逆电对

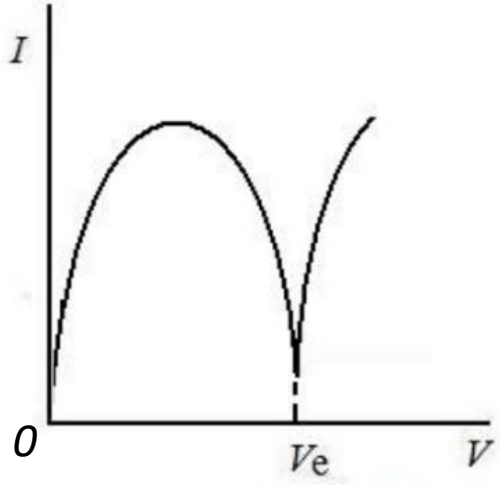
$$Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$$

$Ce^{4+} \rightarrow Fe^{2+}$

Initially the current increases

Near the endpoint the current decreases

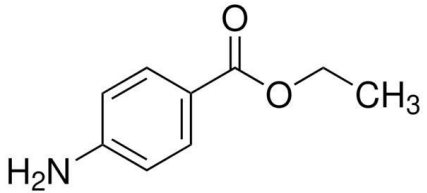
After the endpoint the current increases sharply



Current-change curve for titration of

$$C_9H_{11}NO_2 + NaNO_2 + 2\,HCl \longrightarrow C_9H_{10}ClN_3O_2 + NaCl + 2\,H_2O$$

II. Applications of dead-stop titration



Example: Assay of benzocaine苯佐卡因

Before the endpoint, sodium nitrite亚硝酸钠reacts with benzocaine苯佐卡因by diazotization重氮化反应. There is no reversible couple in the solution, no current passes, and the ammeter needle remains at zero (or near zero). After the endpoint, a slight excess of $NaNO_2$ reacts under acidic conditions to form NO, which forms the reversible HNO_2/NO couple with HNO_2 in the solution. The electrode reactions at the two electrodes are:

阳极Anode: $NO+H_2O \rightarrow HNO_2+H^+ + e$

阴极Cathode: $HNO_2 + H^+ + e \rightarrow NO+H_2O$

At this time, current passes through the circuit, the ammeter needle suddenly deflects and no longer returns to zero, indicating that the titration endpoint has been reached.
电流指针突然偏转不再回零，指示到达滴定终点

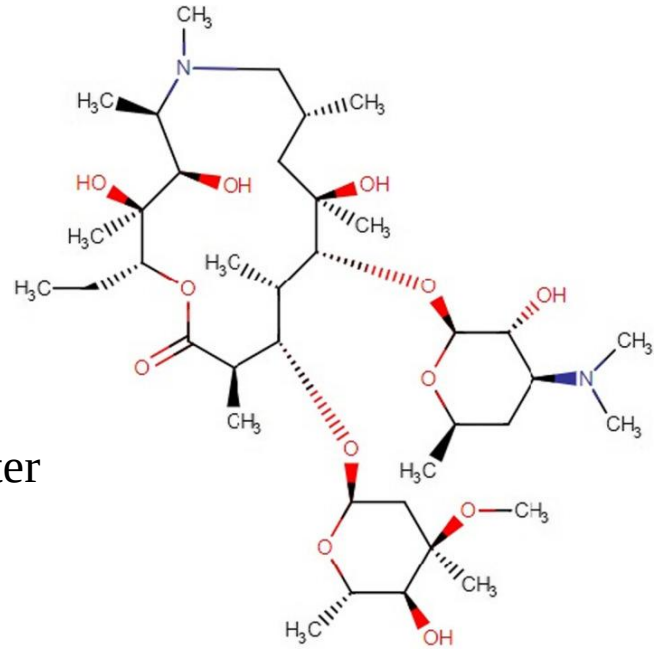
II. Applications of dead-stop titration

Example: Determination of trace water in azithromycin 阿奇霉素for injection

Before the endpoint, there is no reversible couple in the solution and the ammeter needle is at zero. After the endpoint, a slight excess of I_2 forms the reversible I_2/I^- couple with I^- in the solution. The electrode reactions are:

阳极Anode: $2I^- \rightarrow I_2 + 2e$

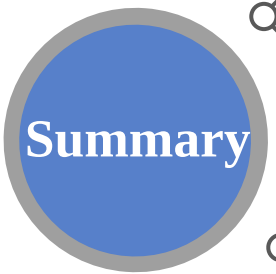
阴极Cathode: $I_2 + 2e \rightarrow 2I^-$



At this time, current passes through the circuit, the ammeter needle suddenly deflects and no longer returns to zero, indicating that the titration endpoint has been reached.

电流指针突然偏转不再回零，指示到达滴定终点

Chapter Summary



1. Basic principles of potentiometry

Structures and principles of galvanic cells and electrolytic cells; concepts and classification of indicator and reference electrodes

2. Direct potentiometry

Principles and methods for measuring solution pH; construction, performance, and precautions of the glass electrode; significance and role of the selectivity coefficient of ion-selective electrodes

3. Potentiometric titration and dead-stop titration

Basic principles and methods for determining the endpoint

Chapter Summary

小结

1.电位法的基本原理

原电池与电解池的结构与原理；指示电极和参比电极的概念及分类

2.直接电位法

溶液pH的测定原理及方法；玻璃电极构造、性能及注意事项；离子选择电极的选择性系数的意义、作用

3.电位滴定法和永停滴定法

基本原理及确定终点的方法

Thank you for watching